

ANNA BODONHELYI

AI Researcher | Data Scientist



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SUMMARY

Machine Learning PhD (Dissertation Submitted) and Applied AI Researcher, combining a deep technical foundation in machine learning with dual Master's degrees in Mechatronics and Artificial Intelligence. I specialize in developing multimodal LLM architectures, federated learning frameworks, and advanced computer vision pipelines. Beyond algorithmic research, I am deeply passionate about human-AI interaction, AI outreach, and building production-ready systems that bridge the gap between complex theoretical models and tangible, real-world applications. My ultimate drive is to build intelligent, embodied systems that actively engage and augment human capabilities.

WORK EXPERIENCE

Machine Learning Researcher | Technical University of Munich

Febr 2023 – Present

- **PhD Dissertation:** AI-Driven Educational Technologies: From Detection to Engagement (Advisor: Prof. Dr. Enkelejda Kasneci) at TUM School of Computation, Information and Technology (CIT)
Munich Center for Machine Learning (MCML) member
- **Applied AI & LLMs:** Architected proactive video assistants and simulated medical communication environments using multimodal AI and Large Language Models (Python, PyTorch).
- **Federated Learning & Privacy:** Engineered novel federated machine learning frameworks to optimize edge-computing training efficiency, alongside privacy-preserving multimodal pipelines for affective computing.
- **Strategy & Leadership:** Serve on the TUM EdTech Center management team, leading the coordination of strategic events, cross-functional projects, and the design of digital advertising and social media initiatives.
- **AI Outreach & Teaching:** Design and lead the hands-on "Creative with AI" (Kreativ mit KI) workshop at Schloss Elmau for diverse stakeholders, and instruct university courses on Human-AI Interaction and machine learning.

Working Student | Siemens Healthcare GmbH

Apr 2022 – Jan 2023

- US Patent (WO2024108104): Systems and Methods for Classifying Blood Cells
 - Engineered an automated computer vision pipeline utilizing GAN-based style transfer and multi-output learning to virtually stain and classify unstained blood cells.
 - Developed and trained deep learning models using Python, PyTorch, and OpenCV, optimizing both image quality metrics (SSIM, LPIPS) and diagnostic accuracy to replace costly chemical staining.

Research Assistant | Helmholtz AI

Sep 2021 – Mar 2022

- Developed a full-stack web application for an interactive MRI viewer, integrating and deploying ML models for automated image analysis.
- Built the backend and frontend infrastructure using Flask, Python, and PyTorch, alongside medical imaging libraries for DICOM visualization.

Trainee | Knorr Bremse

Jun 2019 – Jan 2020

- Designed and simulated Electronic Air Control (EAC) systems utilizing MATLAB and Simulink for model-based development and control logic.
- Automated system testing and data logging pipelines by writing Python scripts and managing performance data with SQLite.

EDUCATION

Master's Degree | Technical University of Munich

Oct 2020 – Feb 2023

- Major in Robotics, Cognition, Intelligence
- Thesis: Classification and Digital Staining of Red Blood Cells with Deep Learning

Master's Degree | Budapest University of Technology and Economics

Feb 2020 – Jan 2022

- Major in Mechatronics Engineering
- Specialization: Intelligent Embedded Mechatronic Systems
- Thesis: Deep Learning-Based Classification of Left Ventricular Hypertrophy

Bachelor's Degree | Budapest University of Technology and Economics

Sep 2016 – Jan 2020

- Major in Mechatronics Engineering
- Thesis: Examination of the Deformation of Vehicle Quarter Glass Injection Moulding Elastomers Caused by Compressed Air (**National Students' Science Conference, 2019 (Second place)**)
- ERASMUS Semester (Oct 2018 – Feb 2019) with Campus Mundi scholarship at the Karlsruhe Institute of Technology

PUBLICATIONS

- **Bodonyi, A.**, Curinier, A., Bühler, B., Anders, G., Rausch, L., Huff, M., et al. (2026). Automatic Mind Wandering Detection in Educational Settings: A Systematic Review and Multimodal Benchmarking. **Computers & Education**
- **Bodonyi, A.***, Wang, M.*, Bozkir, E., Bühler, B., & Kasneci, E. (2026). Safeguarding Privacy: Privacy-Preserving Detection of Mind Wandering and Disengagement Using Federated Learning in Online Education. International Journal of Artificial Intelligence in Education (**IJAIED 2026**) * **These authors contributed equally.**
- Otmani, K., **Bodonyi, A.**, Bühler, B., & Kasneci, E. (2026). Should AI Ask First? Investigating the Effects of Proactive vs. Reactive AI Mentoring in Self-Directed Learning. International Conference on Artificial Intelligence in Education (**AIED 2026**).
- **Bodonyi, A.**, Thaqi, E., Özdel, S., Bozkir, E., & Kasneci, E. (2025). From Passive Watching to Active Learning: Empowering Proactive Participation in Digital Classrooms with AI Video Assistant. Proceedings of the ACM CHI Conference on Human Factors in Computing Systems (**CHI 2025**).
- **Bodonyi, A.**, Stegemann-Philipps, C., Sonanini, A., Herschbach, L., Szep, M., et al. (2025). Modeling Challenging Patient Interactions: LLMs for Medical Communication Training. Under review. arXiv:2503.22250
- **Bodonyi, A.**, Bozkir, E., Yang, S., Kasneci, E., & Kasneci, G. (2024). User Intent Recognition and Satisfaction with Large Language Models: A User Study with ChatGPT. Under review. arXiv:2402.02136
- Wang, M.,* **Bodonyi, A.***, Bozkir, E., & Kasneci, E. (2024). TurboSVM-FL: Boosting Federated Learning through SVM Aggregation for Lazy Clients. Proceedings of the AAAI Conference on Artificial Intelligence (**AAAI 2024**). * These authors contributed equally.

PATENT

- US Patent (WO2024108104): Systems and Methods for Classifying Blood Cells

SCHOLARSHIPS

- bidt – Bavarian Research Institute for Digital Transformation: 2025
- Campus Mundi Scholarship with ERASMUS: 2018

KEY SKILLS

- Applied Machine Learning
- Multimodal AI & LLMs & VLMs
- Computer Vision & GANs
- Model Deployment (MLOps)
- Technical Communication
- Project Management

LANGUAGES

- English: C1
- German: C1
- Hungarian: Native